# EXPERIMENT : 15

AIM:

Experiment to understand the data preprocessing in Data Science.

DESCRIPTION:

Understand the importance of Data preprocessing in data science.

CODE:

```
import numpy as np
import pandas as pd

# Read dataset
df = pd.read_csv('/content/Data.csv')

# Display dataset and information
print(df)
df.info()

# Find the mode of Country
print(df['Country'].mode())
print(df['Country'].mode()[0])
print(type(df['Country'].mode()))

# Fill missing Country values with mode
df['Country'] = df['Country'].fillna(df['Country'].mode()[0])

# Fill missing Age values with median
df['Age'] = df['Age'].fillna(df['Age'].median())

# Fill missing Salary values with rounded mean
df['Salary'] = df['Salary'].fillna(round(df['Salary'].mean()))

# Display updated dataset
print(df)

# One-hot encoding of Country
```

```
country_encoded = pd.get_dummies(df['Country'])
print(country_encoded)

# Combine encoded Country with Age, Salary and Purchased
updated_dataset = pd.concat(
    [country_encoded, df[['Age', 'Salary', 'Purchased']]],
    axis=1
)

# Convert Purchased values into 0 and 1
updated_dataset['Purchased'] = updated_dataset['Purchased'].replace(
    {'No': 0, 'Yes': 1}
)

# Display final dataset
print(updated_dataset)
```

OUTPUT:

```
  Country   Age   Salary Purchased
0   France  44.0  72000.0        No
1    Spain  27.0  48000.0       Yes
2  Germany  30.0  54000.0        No
3    Spain  38.0  61000.0        No
4  Germany  40.0      NaN       Yes
5   France  35.0  58000.0       Yes
6    Spain   NaN  52000.0        No
7   France  48.0  79000.0       Yes
8  Germany  50.0  83000.0        No
9   France  37.0  67000.0       Yes
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   Country    10 non-null     object
 1   Age        9 non-null      float64
 2   Salary     9 non-null      float64
 3   Purchased  10 non-null     object
dtypes: float64(2), object(2)
memory usage: 452.0+ bytes
0    France
Name: Country, dtype: object
```

```
France
<class 'pandas.core.series.Series'>
  Country   Age   Salary Purchased
0   France  44.0  72000.0        No
1    Spain  27.0  48000.0       Yes
2  Germany  30.0  54000.0        No
3    Spain  38.0  61000.0        No
4  Germany  40.0  63778.0       Yes
5   France  35.0  58000.0       Yes
6    Spain  38.0  52000.0        No
7   France  48.0  79000.0       Yes
8  Germany  50.0  83000.0        No
9   France  37.0  67000.0       Yes
  France  Germany  Spain
0   True    False  False
1  False    False   True
2  False     True  False
3  False    False   True
4  False     True  False
5   True    False  False
6  False    False   True
7   True    False  False
8  False     True  False
9   True    False  False
  France  Germany  Spain   Age   Salary  Purchased
0   True    False  False  44.0  72000.0         0
1  False    False   True  27.0  48000.0         1
2  False     True  False  30.0  54000.0         0
3  False    False   True  38.0  61000.0         0
4  False     True  False  40.0  63778.0         1
5   True    False  False  35.0  58000.0         1
6  False    False   True  38.0  52000.0         0
7   True    False  False  48.0  79000.0         1
8  False     True  False  50.0  83000.0         0
9   True    False  False  37.0  67000.0         1
```

## RESULT:

Data preprocessing was successfully performed on the given dataset. Missing values, duplicate records, and inconsistent data were handled to improve data quality and prepare the dataset for further analysis.